

Structural results for the order-six Lyapunov norm problem

Abstract

We collect partial results for the symmetric-maximizer conjecture in order six. The principal inequality is $\sigma_2(\mathcal{L}_A|_{\mathcal{K}_6}) \leq \|\mathcal{L}_A|_{\mathcal{S}_6}\|_{F \rightarrow F}$: at most one skew singular value can exceed the symmetric norm. We prove exclusions based on common symmetries and record two six-dimensional geometric constructions, including an orthogonal $4 + 2$ polar decomposition and an exact coupling identity. All stated results have proofs; the largest-singular-value comparison is not established.

1 Setting and singular pairs

For $A \in \mathbb{R}^{6 \times 6}$, let $\mathcal{L}_A(X) = AX + XA^T$ and equip matrix spaces with $\langle X, Y \rangle_F = \text{tr}(X^T Y)$. All operator norms are induced by the Frobenius norm. Write

$$\mathcal{S}_6 = \{X : X^T = X\}, \quad \mathcal{K}_6 = \{X : X^T = -X\}, \quad s = s(A) = \|\mathcal{L}_A|_{\mathcal{S}_6}\|_{F \rightarrow F}.$$

Let $\sigma_1 \geq \dots \geq \sigma_{15} \geq 0$ be the singular values of $\mathcal{L}_A|_{\mathcal{K}_6}$, counted with multiplicity. The target is $\sigma_1 \leq s$. Chen–Tian proved the comparison for orders at most five [1]; Kressner–Vandereycken gave counterexamples in every order at least seven [2]. This note makes no priority claim for the partial results.

Both parity spaces are invariant under $\mathcal{L}_A$ and its adjoint $\mathcal{L}_A^* = \mathcal{L}_{A^T}$. Compactness of the skew unit sphere and the Rayleigh principle therefore give, when $\sigma = \sigma_1 > 0$,

$$K, N \in \mathcal{K}_6, \quad \|K\|_F = \|N\|_F = 1, \quad \mathcal{L}_A(K) = \sigma N, \quad \mathcal{L}_{A^T}(N) = \sigma K. \quad (1)$$

Indeed, choose a unit maximizer K and set $N = \mathcal{L}_A(K)/\sigma$; $\mathcal{L}_{A^T}\mathcal{L}_A(K) = \sigma^2 K$ gives the second equation. No projection is needed because the adjoint preserves skew-symmetry. If $\sigma = 0$, the desired comparison is immediate.

For symmetric P, Q and skew K, N , cyclicity of the trace gives

$$\langle Q, \mathcal{L}_A(P) \rangle_F = 2 \text{tr}(APQ), \quad \langle N, \mathcal{L}_A(K) \rangle_F = -2 \text{tr}(AKN). \quad (2)$$

Also, for $U = P + K$,

$$\text{tr}(U^2) = \|P\|_F^2 - \|K\|_F^2, \quad \text{tr}(\mathcal{L}_A(U)^2) = \|\mathcal{L}_A(P)\|_F^2 - \|\mathcal{L}_A(K)\|_F^2. \quad (3)$$

All constructions below preserve the original Frobenius metric.

2 Rank-four companions and spectral isolation

Lemma 1 (Rank-four companion). *For every $A \in \mathbb{R}^{6 \times 6}$ and $Z \in \mathcal{K}_6$ with $\text{rank } Z \leq 4$, there is $P \in \mathcal{S}_6$ such that*

$$P^2 = -Z^2, \quad \|\mathcal{L}_A(P)\|_F \geq \|\mathcal{L}_A(Z)\|_F.$$

Consequently, $\|\mathcal{L}_A(Z)\|_F \leq s\|Z\|_F$.

Proof. By orthogonal similarity, put

$$Z = \text{diag}(k_1 J_0, k_2 J_0, 0_2), \quad J_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Independent orthogonal changes within the first two planes diagonalize their cross block of A by a singular value decomposition. Reflections only change the signs of the k_i , which may be zero. Thus we may assume $a_{14} = a_{23} = 0$. Set

$$P = \text{diag}(k_1 F_0, k_2 F_0, 0_2), \quad F_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then $P^2 = -Z^2$. To check the output difference including all outside couplings, factor

$$U = P + Z = XY^T, \quad X = [2k_1 e_1, 2k_2 e_3], \quad Y = [e_2, e_4].$$

Since $Y^T X = 0$, expansion gives $\text{tr}(\mathcal{L}_A(U)^2) = \text{tr}((Y^T A X)^2) + \text{tr}((Y^T A^T X)^2)$. Using the two zero entries, this equals

$$4k_1^2(a_{12}^2 + a_{21}^2) + 4k_2^2(a_{34}^2 + a_{43}^2) \geq 0. \quad (4)$$

Equation (3) completes the proof. No invariance of the support of Z under A was assumed. $\square$

Theorem 2 (Spectral isolation). *For every $A \in \mathbb{R}^{6 \times 6}$,*

$$\boxed{\sigma_2(\mathcal{L}_A|_{\mathcal{K}_6}) \leq \|\mathcal{L}_A|_{\mathcal{S}_6}\|_{F \rightarrow F}}. \quad (5)$$

If $\sigma_1 > s$, then σ_1 is simple and both members of its skew singular pair are invertible.

Proof. Every real two-dimensional subspace of $\mathcal{K}_6$ contains a nonzero singular matrix. In fact, for an orthonormal basis Z_1, Z_2 , the continuous function

$$f(\theta) = \text{pf}(\cos \theta Z_1 + \sin \theta Z_2)$$

satisfies $f(\theta + \pi) = -f(\theta)$ because the order-six Pfaffian is homogeneous of degree three. Hence it vanishes somewhere; the corresponding unit matrix has rank at most four.

If $\sigma_2 > s$, every nonzero matrix in the span of two top skew right singular matrices has Rayleigh quotient greater than s^2 . The preceding rank-four matrix contradicts Lemma 1. Simplicity follows immediately. A singular K is excluded by that lemma; a singular N is excluded by applying it to A^T , whose restricted norms are the same. $\square$

3 Exclusion by common symmetries

An *orthogonal complex structure* is a real matrix J with $J^T = -J$ and $J^2 = -I$. A *symmetric orthogonal involution* is a real matrix H with $H^T = H$ and $H^2 = I$.

Proposition 3 (Exact symmetry witnesses). *For a pair (1), either a common commuting orthogonal complex structure J or a common anticommuting symmetric orthogonal involution H proves $s \geq \sigma$. More precisely, with $T = J$ or $T = H$, respectively,*

$$s^2 \geq \sigma^2 + \max\{\|[A, T]K\|_F^2, \|[A^T, T]N\|_F^2\}. \quad (6)$$

Proof. In either case $P = TK$ and $Q = TN$ are symmetric unit matrices and $PQ = -KN$. Thus (2) gives $\langle Q, \mathcal{L}_A(P) \rangle_F = \sigma$. On the other hand,

$$\mathcal{L}_A(P) = \sigma Q + [A, T]K.$$

The two terms are orthogonal, proving the first squared bound. Apply the same argument to A^T, N, K for the second. If K is invertible, equality $s = \sigma$ forces $[A, T] = 0$. $\square$

Theorem 4 (Joint-commutant exclusion). *A pair (1) proves $s \geq \sigma$ if either*

$$\{R : RK = KR, RN = NR\} \neq \mathbb{R}I$$

or there is a nonzero real symmetric H with $HK = -KH$ and $HN = -NH$.

Proof. Singular K or N is covered by Lemma 1, so assume both invertible. If a nonscalar symmetric matrix commutes with K, N , a proper spectral projection gives a common orthogonal splitting $K = K_1 \oplus K_2$, $N = N_1 \oplus N_2$. Invertibility and skew-symmetry force block sizes $2 + 4$ or $4 + 2$. Write $\widehat{K}_i$ for K_i embedded by zeros. The diagonal blocks of (1) and Lemma 1 imply

$$\sigma^2 \|N_i\|_F^2 \leq \|\mathcal{L}_A(\widehat{K}_i)\|_F^2 \leq s^2 \|K_i\|_F^2.$$

Summing proves $\sigma \leq s$; off-diagonal blocks of A need not vanish.

The joint commutant is closed under transpose. Thus any nonscalar member either has a nonscalar symmetric part or has a nonzero skew part W . In the latter case $-W^2$ is a common symmetric commuting matrix. If it is nonscalar, use the preceding argument; otherwise $-W^2 = c^2 I$, $c > 0$, and $J = W/c$ is covered by Proposition 3.

Finally, a symmetric common anticommuting $H \neq 0$ has H^2 in the joint commutant. If H^2 is nonscalar, the first part applies. Otherwise $H^2 = c^2 I$, $c > 0$, and H/c is covered by Proposition 3. $\square$

4 A six-dimensional simultaneous reflection

Theorem 5 (Three-coordinate reflection). *For any orthogonal complex structure J and any $N \in \mathcal{K}_6$, there exists a symmetric orthogonal involution H such that*

$$HJ = -JH, \quad HN = -NH.$$

Its two eigenspaces both have dimension three.

Proof. Identify $\mathbb{R}^6$ orthogonally with $\mathbb{C}^3$ so that $Jz = iz$. A real skew map has the form $Nz = iEz + F\bar{z}$, with $E = E^*$ and $F^T = -F$. A unitary change of coordinates makes E real diagonal. Seek

$$Hz = V\bar{z}, \quad V = \text{diag}(v_1, v_2, v_3), \quad |v_i| = 1.$$

This is a real symmetric orthogonal involution anticommuting with J . Anticommutation with N reduces to $V\bar{F} + F\bar{V} = 0$, or

$$v_i v_j = \rho_{ij} := -f_{ij}/\bar{f}_{ij} \quad (f_{ij} \neq 0).$$

Assign arbitrary unit phases at zero entries and choose

$$v_1^2 = \frac{\rho_{12}\rho_{13}}{\rho_{23}}, \quad v_2 = \frac{\rho_{12}}{v_1}, \quad v_3 = \frac{\rho_{13}}{v_1}.$$

All three equations hold. Finally J interchanges the two eigenspaces of H . Repeated eigenvalues of E and zero entries of F are included. $\square$

In particular, $K^2 = -a^2 I$, $a > 0$, for either member of a maximizing pair implies $s \geq \sigma$: apply the theorem to $J = K/a$ and then Proposition 3 (or use the adjoint pair). For unequal weights, anticommutation with the polar structure of K does *not* imply anticommutation with K .

The reflection theorem fails in real dimension eight. On $\mathbb{C}^4$, take $Jz = iz$ and $Nz = i\Lambda z + F\bar{z}$, where $\Lambda = \text{diag}(10, 20, 30, 40)$ and the only nonzero upper-triangular entries of F are $f_{12} = f_{13} =$

$f_{24} = 1, f_{34} = i$. A putative reflection has the form $V\bar{z}$; anticommutation with $i\Lambda$ forces V to be diagonal. The equations would require

$$v_1v_2 = v_1v_3 = v_2v_4 = -1, \quad v_3v_4 = 1,$$

contradicting equality of the products $(v_1v_2)(v_3v_4) = (v_1v_3)(v_2v_4)$.

5 Polar 4 + 2 geometry and the coupling identity

For invertible skew K, N , define their polar complex structures

$$J = K(-K^2)^{-1/2}, \quad L = N(-N^2)^{-1/2}.$$

They commute with K, N , respectively, and $-JK \succ 0, -LN \succ 0$. If $L = \pm J$, the common commuting structure already proves the norm comparison for a singular pair. The following treats the other case.

Theorem 6 (Common polar two-plane). *Let J, L be orthogonal complex structures on $\mathbb{R}^6$, and set*

$$S = -\frac{1}{2}(JL + LJ), \quad \varepsilon = \text{pf}(J)\text{pf}(L) \in \{\pm 1\}.$$

For some $c \in [-1, 1]$,

$$\det(tI - S) = (t - \varepsilon)^2(t - c)^4, \quad c = \frac{\text{tr } S - 2\varepsilon}{4}. \quad (7)$$

If $L \neq \varepsilon J$, then $c \neq \varepsilon$ and $\Pi_E = (S - cI)/(\varepsilon - c)$ projects orthogonally onto a common invariant two-plane E . On E , $L = \varepsilon J$; on $F = E^\perp$,

$$J_F L_F + L_F J_F = -2cI_F. \quad (8)$$

Proof. The matrix S is symmetric, commutes with J, L , and has spectrum in $[-1, 1]$, as follows from $(J \mp L)^T(J \mp L) = 2(I \mp S)$. On an eigenspace with eigenvalue $c \in (-1, 1)$, put $M = (L - cJ)/\sqrt{1 - c^2}$. Then $M^2 = -I$, $M^T = -M$, and $JM = -MJ$. For any unit x in this eigenspace, x, Jx, Mx, JMx form an orthonormal invariant four-frame; iteration shows that this eigenspace has dimension divisible by four. At an endpoint, $L = J$ or $L = -J$, and the eigenspace has even dimension. Thus a nonproportional pair in dimension six splits as 4 + 2.

On the four-dimensional block the Pfaffians of J_F, L_F have the same sign: for an interior block use the path $\cos \theta J_F + \sin \theta M$; for an endpoint block use $\text{pf}(-J_F) = \text{pf}(J_F)$. Hence the relative sign on the two-plane equals ε . This proves (7) and the projector formula. The proportional cases give $S = \varepsilon I$ directly. $\square$

Proposition 7 (Constrained coupling blocks). *Assume $L \neq \pm J$ and use the orthogonal splitting $F \oplus E$ from Theorem 6. With $J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$,*

$$K = \begin{pmatrix} K_F & C \\ -C^T & kJ_2 \end{pmatrix}, \quad N = \begin{pmatrix} N_F & D \\ -D^T & \varepsilon nJ_2 \end{pmatrix}, \quad k, n > 0. \quad (9)$$

Here K_F, N_F are invertible skew matrices of order four, and

$$J_F C = C J_2, \quad L_F D = \varepsilon D J_2, \quad C^T C = \alpha^2 I_2, \quad D^T D = \beta^2 I_2 \quad (10)$$

for $\alpha, \beta \geq 0$. Their overlap satisfies

$$\|C^T D\|_F^2 - 2\varepsilon c \det(C^T D) = \alpha^2 \beta^2 (1 - c^2). \quad (11)$$

Proof. Commutation of K with J and of N with L gives the first two relations in (10). Hence $C^T C$ and $D^T D$ are symmetric matrices commuting with J_2 , so are scalar. Positivity of $-JK$ and $-LN$ on their principal blocks proves $k, n > 0$ and invertibility of K_F, N_F .

For (11), first suppose $\alpha\beta > 0$ and $|c| < 1$. Put $s_0 = \sqrt{1 - c^2}$ and $M = (L_F - cJ_F)/s_0$. By the intertwining relations, choose unit $x, y \in F$ with

$$C/\alpha = [x, J_F x], \quad D/\beta = [y, \varepsilon L_F y].$$

Expand $y = ax + bJ_F x + dMx + eJ_F Mx$ in the orthonormal four-frame, so $a^2 + b^2 + d^2 + e^2 = 1$. Then

$$G := \frac{C^T D}{\alpha\beta} = \begin{pmatrix} a & -\varepsilon(cb + s_0 d) \\ b & \varepsilon(ca + s_0 e) \end{pmatrix}.$$

Expansion gives $\|G\|_F^2 - 2\varepsilon c \det G = 1 - c^2$. If a coupling vanishes, the identity is immediate. At $c = \pm 1$, one has $L_F = cJ_F$; the same computation without the s_0 terms gives $\|G\|_F^2 = 2\varepsilon c \det G$. Thus the endpoint cases are included. $\square$

For a unit pair the exact norm budgets are

$$\|K_F\|_F^2 + 4\alpha^2 + 2k^2 = 1, \quad \|N_F\|_F^2 + 4\beta^2 + 2n^2 = 1. \quad (12)$$

Equations (9)–(12) require no maximizing assumption. They describe the geometry of arbitrary invertible skew pairs. The two-plane reduces their *polar structures*, not necessarily K, N : the coupling blocks must not be discarded.

6 What remains to be proved

Combining Theorems 2 and 4, any strict counterexample must have

$$\begin{aligned} \sigma_1 > s \geq \sigma_2, \quad \det K \det N \neq 0, \\ \{R : RK = KR, RN = NR\} = \mathbb{R}I, \\ \{H = H^T : HK = -KH, HN = -NH\} = \{0\}. \end{aligned} \quad (13)$$

The reflection theorem also excludes equal positive canonical weights for either singular matrix. The polar structures of a counterexample cannot be proportional, so the coupled form (9) applies. The unresolved issue is a norm-controlled treatment of those couplings. In particular,

$$-KN = \begin{pmatrix} -K_F N_F + CD^T & -K_F D - \varepsilon n C J_2 \\ C^T N_F + k J_2 D^T & C^T D + \varepsilon k n I_2 \end{pmatrix}.$$

Separate order-four and order-two companions do not control these mixed terms. Identity (11) restricts their geometry, but no inequality proved here converts it into $s \geq \sigma_1$. Thus the main established conclusion is *spectral isolation and structural exclusion*, not a proof of the full order-six conjecture. No numerical evidence or unproved optimization claim is used in the results above.

References

- [1] S. Chen and Y. Tian, *Note on “On the Singular ‘Vectors’ of the Lyapunov Operator” by R. Byers and S. Nash*, SIAM J. Matrix Anal. Appl. **36** (2015), 1069–1072. doi:10.1137/140974031.
- [2] D. Kressner and B. Vandereycken, *A counterexample to the symmetric-maximizer conjecture for Lyapunov operators*, arXiv:2608.20875v1 (2026). arXiv:2608.20875v1.